A locally toroidal polytope with a nonorientable vertex figure

Human Compression project

Fernandes, Leemans and Ivić Weiss [1], pp. 18–19, mention the polytope $\{\{4,4\}_{(4,0)},\{4,6\}_3\}$ and state that its group has order 768 and its vertex figure has nonorientable genus four. They explicitly leave further study of this example aside. We verify these assertions separately from their tables of orientably regular and chiral examples. Here the subscript in $\{4,6\}_3$ is the Petrie length.

Proposition 1. *Let U be the group generated by involutions a,b,c,d with*

$$(ab)^4=(bc)^4=(cd)^6=(ac)^2=(ad)^2=(bd)^2=1,\qquad (abcb)^4=(bcd)^3=1.$$

Then U, with this ordered generating set, is a string C-group of order 768. Its facets are $\{4,4\}_{(4,0)}$, and its vertex figures are the nonorientable maps $\{4,6\}_3$ of genus four. It is universal for these two prescribed local presentations.

Proof. The program `verify.g` forms the displayed finitely presented group itself. GAP's exact finite group algorithms give $|U|=768$ and a bijective homomorphism to a permutation group of degree 16. For clarity, the images of the four distinguished generators are

$$\begin{aligned}
a&=(1\,2)(3\,6)(4\,7)(5\,8)(9\,11)(10\,12)(13\,14)(15\,16),\\
b&=(1\,3)(2\,6)(4\,5)(7\,8)(9\,13)(10\,14)(11\,15)(12\,16),\\
c&=(1\,4)(2\,7)(3\,9)(5\,10)(6\,11)(8\,12)(13\,14)(15\,16),\\
d&=(1\,5)(2\,8)(3\,4)(6\,7)(9\,12)(10\,11)(13\,16)(14\,15).
\end{aligned}$$

This permutation image alone would give only a lower bound on the universal group. The calculation of the finitely presented group supplies the upper bound and checks bijectivity. For every pair of subsets $I,J\subseteq\{a,b,c,d\}$, the program checks

$$\langle I\rangle\cap\langle J\rangle=\langle I\cap J\rangle.$$

These are all 256 pairs, including the empty and full subsets. The adjacent product orders are exactly $4,4,6$. The standard coset construction for string C-groups therefore gives a regular abstract polytope with this group and marking.

The facet and vertex stabilizers $\langle a,b,c\rangle$ and $\langle b,c,d\rangle$ have orders 128 and 48. Independently forming their three-generator local presentations gives these same orders; the maps sending local generators to the displayed distinguished generators are checked bijective. Thus there are no additional local relations. The first presentation is the square torus $\{4,4\}_{(4,0)}$: in the affine square reflection group, *abcb* is a unit translation, and its fourth power and its conjugates generate the period lattice $4\mathbb{Z}^2$. The quotient has order $8\cdot16=128$. The second is exactly the Petrie quotient $\{4,6\}_3$; the product *bcd* has order three.

In the vertex figure the numbers of vertices, edges and faces are respectively

$$[\langle b,c,d\rangle:\langle c,d\rangle]=4,\qquad [\langle b,c,d\rangle:\langle b,d\rangle]=12,\qquad [\langle b,c,d\rangle:\langle b,c\rangle]=6.$$

The Euler characteristic is therefore $4-12+6=-2$. The relation $(bcd)^3=1$ is a closed walk of odd length nine in its flag graph. Orientability would assign opposite orientations to adjacent

flag triangles and hence make this graph bipartite. The odd closed walk rules this out. The map is a connected closed nonorientable surface, so its genus is $2 - (-2) = 4$.

Finally, any regular polytope with these local presentations satisfies the displayed defining relations. Its distinguished generators therefore receive a surjective homomorphism from U. Since U itself has the required intersection property and local groups, it is the universal object for these local presentations. □

The group order and genus are assertions of the cited source, not new claims of discovery. The new work is their explicit verification in the present proof system. The finite presentation, all subgroup intersections and the local isomorphisms are required exact computations in GAP; this note is not a complete Lean proof. It settles this isolated source assertion and does not answer any of that paper's open questions.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the retained 21-page 2018 edition, pp. 18–19.